Decent Coaching Classes

Std. X (SB) — Mind Your Math - I

Answer Key — Part I (Ch 1, 2, 3.1) & Part II (Ch 1, 2, 7)

Q.1 (A)

(1) (a) $ax + by + c = 0$

(2) (b) $2x^2 - 3x + 1 = 0$

(3) (b) 9 : 25

(4) (b) (5, 12, 13)

Q.1 (B)

(1) Volume of a cone = $(1/3)\pi r^2 h$

(2) Common difference $d = 5$

Q.2

(1) $x + y = 7$ and $x - y = 1$. Adding: $2x = 8 \Rightarrow x = 4$, and $y = 3$.

(2) $x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0 \Rightarrow x = 2$ or $x = 3$

(3) Yes, it is an A.P. (differences $6-3=3$, $9-6=3$, $12-9=3$ are constant). Common difference $d = 3$

(4) Ratio of areas = (ratio of sides)^2 = $(4/7)^2 = 16/49$

(5) $9^2 + 12^2 = 81 + 144 = 225 = 15^2$. Since it satisfies $a^2 + b^2 = c^2$, (9, 12, 15) is a Pythagorean triplet.

Q.3

(1) Let ten's digit = x, unit's digit = y. Number = $10x + y$. Given $x + y = 9$. Interchanged number = $10y + x$, and $(10y+x) - (10x+y) = 27 \Rightarrow 9(y-x) = 27 \Rightarrow y - x = 3$. Solving $x+y=9$ and $y-x=3$: $y = 6$, $x = 3$. The number = $10(3) + 6 = 36$.

(2) $2x^2 + 3x - 2 = 0$. Using the formula, $x = [-3 \pm \sqrt{(9+16)}]/4 = [-3 \pm 5]/4$. So $x = 1/2$ or $x = -2$

(3) Hypotenuse = $\sqrt{(9^2 + 12^2)} = \sqrt{(81+144)} = \sqrt{225} = 15$ cm

(4) Volume = $\pi r^2 h = (22/7) \times 7^2 \times 10 = (22/7) \times 49 \times 10 = 1540\ cm^3$

Q.4

(1) By the Basic Proportionality Theorem, AD/DB = AE/EC. So $6/4 = 9/EC \Rightarrow EC = 6$ cm. Since $\Delta ADE \sim \Delta ABC$ (AA test, as DE || BC), AB = AD + DB = 10 cm, so AD/AB = 6/10 = 3/5. Hence $A(\Delta ADE) : A(\Delta ABC) = (3/5)^2 =$ 9 : 25 (ratio of areas of similar triangles = ratio of squares of corresponding sides).

(2) Volume of cylinder = $\pi r^2 h = (22/7) \times 6^2 \times 14 = 1584\ cm^3$. Volume of cone = $(1/3) \times 1584 = 528\ cm^3$. Since the cone's capacity ($528\ cm^3$) is much less than the cylinder's water ($1584\ cm^3$), all the water will NOT fit. Volume of water that overflows = $1584 - 528 = 1056\ cm^3$